

[CH 7] Real Time: Reaping the Whirlwind

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- In many ways, early computers were not much better than their predecessors and were being used unimaginatively
- They were basically electronic counterparts of the punched card accounting machines. They were not even necessarily more efficient or cheaper
- Computers made a real difference in real-time systems (a system that responds to messages in "real time" = seconds, sometimes even faster)
- They were the first office machines able to achieve such speed

Project Whirlwind

- Aircraft trainers were simulators used to train pilots far cheaper and more safely than if they were training on an actual plane. However, **a new simulator had to be built for every plane type**
- The army asked **MIT's Servomechanisms Lab** to design an universal simulator that could be used for training for any plane
- Jay W. Forrester was put in charge of the project

Jay W Forrester

- 26 when he was put in charge of the project in 1944
- Whirlwind went wildly over-budget and over-time:
 - Cost \$8 Million
 - Took 8 years to develop
- Project Whirlwind ended up establishing real-time computing, but this was unforeseeable at the start
- Forrester recruited a team. His assistant was MIT graduate student Robert R. Everett
- Forrester dealt with the mechanical elements of the trainer quite easily, but the computing part seemed impossible: analog computers would simply not be fast enough
- Perry Crawford, MIT graduate student, told him about digital computers for the first time in summer 1945
- During the Moore School lectures of 1946, Everett was enrolled as a student and Crawford gave a lecture on real-time application of digital computing
- By 1946 Forrester was convinced that they should use digital computers, even though it would be far more expensive
- With increased funding from the Bureau of Aeronautics, they set out to build a digital computer. 1945
- They found reliable memory to be the most difficult task
 - The mercury delay line was far too slow. Other computers were doing 1000 - 10,000 operations/s, while Whirlwind would need 100,000
 - Electrostatic storage seemed promising, but it was still a "blue sky idea"

Project SAGE

- During the cold war, a need to modernise the US air defense system became apparent
- George E. Valley, professor of physics at MIT, was the chairman of a committee (ADSEC or Air Defense System Engineering Committee) that was in charge of dealing with this issue. Among his recommendations was the inclusion of a computer-based command-and-control system
- Despite hearing negative information about the Whirlwind, he went and visited it to see for himself
- The machine had just started running test programs and Valley chose the team and the project

- The support from Valley and the Airforce made the criticism that Forrester was receiving die down and his requested funding for 1951 was eventually fulfilled
- Months later, project Lincoln started at MIT to design the SAGE system (Semi-Automatic Ground Environment). Whirlwind was part of project Lincoln.
- End of 1951, Bill Papian demonstrated core-memory, which replaced the unreliable and low-capacity electrostatic memory in the Whirlwind, making it the fastest computer in the world by far
- In 5 years, core memory had replaced all other types of memory and, although other people were working on the same idea at the same time, the patent got MIT millions of dollars
- The core memory was the last piece of the system. With R&D complete, production had to begin. The technology was transferred to IBM, which turned it into the IBM AN/FSQ-7
- The final SAGE system consisted of a network of 21 Direction Centers, each with two IBM AN/FSQ-7 computer (for reliability, one was operating live and the other was on standby)
- By the time SAGE was militarily ready in 1963, bombers were not the main threat anymore: ICBM's were
- The real contribution of SAGE was not to defense, but to the field of civilian computing. An entire subindustry came together to develop the technologies and implement the hardware, software and networking
 - These contractors were IBM, Burroughs, Bell Laboratories and many others
 - Technologies included: printed circuit boards, core memories, mass-storage devices, CRT based graphical displays (which created a culture of interactive computing that the world did not catch up with till the 1970s)
 - SAGE was a training ground for software engineers. "Chances were reasonably high that on a large data-processing job in the 1970s, you would find someone that had worked on SAGE"
- For IBM, SAGE accounted for half a billion dollars throughout the 1950s. 20% of the IBM workforce was working on it at some point and the company got a lead in processor technology, mass-storage devices, real-time systems which it put to commercial use and never lost

SABRE: a Revolution in Airline Reservations

The first project to draw from IBM's know how and the innovations derived from SAGE

- SABRE transformed airline reservations
- Before SABRE, in the 1940s and 50s, reservation offices were massive operations with tens of clerks manually processing thousands of calls a day from customers for reservations, information, cancellations or ticket purchases
- This was fine until the mid 40s, but once airline travel started increasing, the clerical system started nearing its maximum capacity
- Airlines did not need the batch-processing that most computers were using. They needed real-time computing
- American Airlines commissioned equipment manufacturer Teleregister to create an electromechanical system for this problem
 - The resulting system was the Reservisor
 - It dealt with sales but was not integrated into the information system. It just kept track of occupied and free seats on planes
 - The information system was still updated manually => discrepancies arose => the system was unreliable
 - The Magnetronic Reservisor was an improved version of the original, but still left record keeping and ticket issuing to be done manually
- By 1953, American Airlines had reached a crisis: reservation costs had become insupportable and new, expensive jet airliners and increased competition would make the problem even worse. Moreover, with AA planning to buy 30 Boeing 707s with 112 passengers each, that could do transcontinental flights in 6 hours, it would exceed the message sending speed of the current

reservations system

- Thomas Watson Jr. had determined that IBM would enter the markets of airline reservations, department stores, banks, railroads, etc.
- 1952 IBM hired Perry Crawford
- In 1954 a joint study team was created to deal with the reservations problem
 - They concluded that the ideal electronic solution was still years in the future
 - They recommended a punched card system in the meantime
 - During 1955-1959, an improved reservisor and a new punched-card system were in operation
- IBM and AA continued working towards the target of an electronic computer. It would need to be a solid state computer with core memory and a large random access disk storage unit.
- Spring 1960 the complete specification was presented to AA management for a cost of just under \$40 million
- The project was implemented in 1960-1963 and involved 200 technical personnel writing 1 million lines of code
 - Largest civilian computerization project up to that point
- The central reservation system was based on 2 IBM 7090 mainframes duplexed for reliability
 - These were connected to 16 disk storage units (total capacity 800 million characters) => largest online storage
 - Connected to 1,100 agents in 50 cities through 10,000 miles of telecommunications lines
 - Could handle close to 10 million passenger reservations per year:
 - 85,000 daily telephone calls, 30,00 daily requests for fare quotations, 40,000 daily passenger reservations, 30,000 daily queries to and from other airlines, 20,000 daily ticket sales
 - Most transactions were handled in 3 seconds
 - It provided far improved passenger service
 - Operators could see passenger's telephone contacts, meal preference, car and hotel reservations, etc. instead of just the number of seats on the plane
 - The system was so good that it gradually took over the airline's entire operation (crew scheduling, flight planning, fuel management, air freight, etc)
 - It was fully operational in 1964
- During 1960 and 1961 Delta and Pan American signed up with IBM and their systems became operational in 1965.
- Eastern Airlines got their system in 1968
- By the 1970s all major carriers had reliable real time systems and communications networks (**not all developed by IBM!**)
- The computerization of airlines was the pioneering act that caused other traditional businesses to follow

The VISA system and Automated Teller Machines (ATMs)

The 2 roots of the financial revolution of cards and ATMs were:

- Cashless payment cards
- Cash dispensing machines
- The first independent payment-card organization was the Diners Club founded in 1949 by 2 New York entrepreneurs. The Diners Club card could be presented as payment and the owner would be billed monthly

- Barclays was the first British bank to launch a computer center in 1961, contracted with De La Rue.
- 1967 the De La Rue Automatic Cash System (DACS) was put into operation in North London (first in either Europe or the US). It used punched-card vouchers that could be purchased inside the bank. These cards and an authorization number would be entered into the machine and cash would be dispensed
- September 1969, the Chemical Bank was the first US bank to deploy a cash dispenser in Long Island, New York
 - It used the now universal Magstripe containing the card's basic info (invented by IBM about a decade earlier)
- 1971 the chemical bank opened a subsequent model that was a truly automatic teller:
 - Make deposits, check balances, transfer funds between savings and checking accounts, etc
- VISA was the creation of financial visionary Dee Hock. It continues the theme of technologies designed for more efficient monetary transactions
- In the 1950s Bank of America introduced the BankAmericard credit card
 - The main problem was enforcing the credit limit. Payments were allowed up to a ceiling amount and authorization was obtained by telephone for larger amounts
 - Settlement and clearing were slow and led to problems with fraud
- After years of a Bank of America monopoly, other banks decided to shake them off and form the National BankAmericard Incorporated (NBI) in 1970
- NBI and their international branch Ibanco were renamed Visa USA and Visa International in 1976
- For its first 14 years, Visa was headed by Dee Hock
- Hock wanted any value, be it credit, checking, savings or brokerage money, to be available for electronic-based purchases or transfers worldwide and at any time
- The greatest problem was authorization, which could take as long as an hour.
- Hock wanted to entirely automate this by using computers
- NBI/Visa started developing the computers Base I (for authorization) and Base II (for clearing and settlement)
- Thompson Ramo Wooldridge, Inc. (TRW) and McKinsey & Co. ran the development effort for BASE I.
- Base I used a DEC PDP-11/45 computer and provided real-time authorizations in seconds
- Unlike SAGE and SABRE, BASE I was delivered on time in 1973 and within budget
- Base II would be a powerful IBM 370/145. Each of the 88 processing centers would have a DEC PDP-11/10 computer communicating with BASE II
- BASE II stayed within its 7 million USD budget and was delivered on time in November 1974
- In the 1980s Visa added a magstripe and inexpensive Point-of-Sale technology connected directly to BASE I
- 1975 Visa introduced Debit Cards linked to the cardholder's bank account named Visa Debit
- In the mid 1980s Debit cards were linked to the ATM networks
- ATMs initially processed transactions in batch overnight, but gradually they started coming online and updated accounts instantly

Universal Product Code

- In the mid 1960s real-time computing started picking up speed
- The UPC is the barcode now almost universally found on food products manufactured or packaged for retail
- With the number of supermarkets exploding and profit margins getting as low as 1% by 1970, the industry needed to cut costs by raising productivity.
- Checkout counter for about 25% of operating costs and was exasperatingly slow
- A new generation of PoS terminals offered the possibility of automatic data capture for pricing and stock control

- This required product codes to be applied to merchandise
- This was not feasible because of the large variety of products and the low average values of them. It would be too expensive to individually apply the codes to each one
- The implementation of the Universal Product Code was a huge cooperative effort in the food industry
- They agreed upon a 10 digit code (5 for the manufacturer, 5 for the product)
- They decided to use the barcode system, largely developed by IBM, because it was simply the cheapest and most trouble free
- The first barcoded goods started appearing in 1974
- Adoption was easier for manufacturers (who just had to add barcodes onto their labels) than for retailers (who had to buy scanners) but by the mid 1980s most food chains had automated their larger stores' checkout
- The use of barcodes spread rapidly (ex: ISBN barcodes on books) to most retail products